

Multimedia Datenbanken

Colors & Color Models

Prof. Dr. Michael Granitzer
Prof. Dr. Harald Kosch

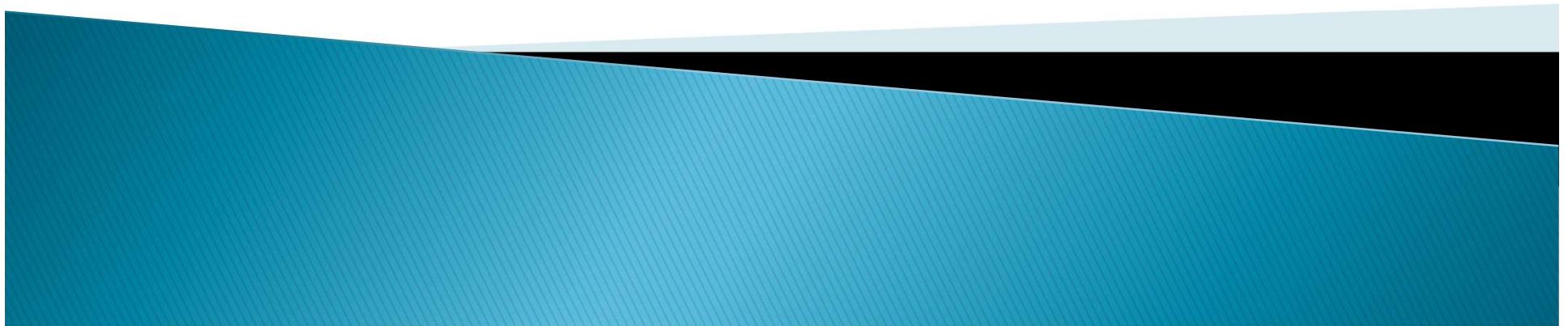


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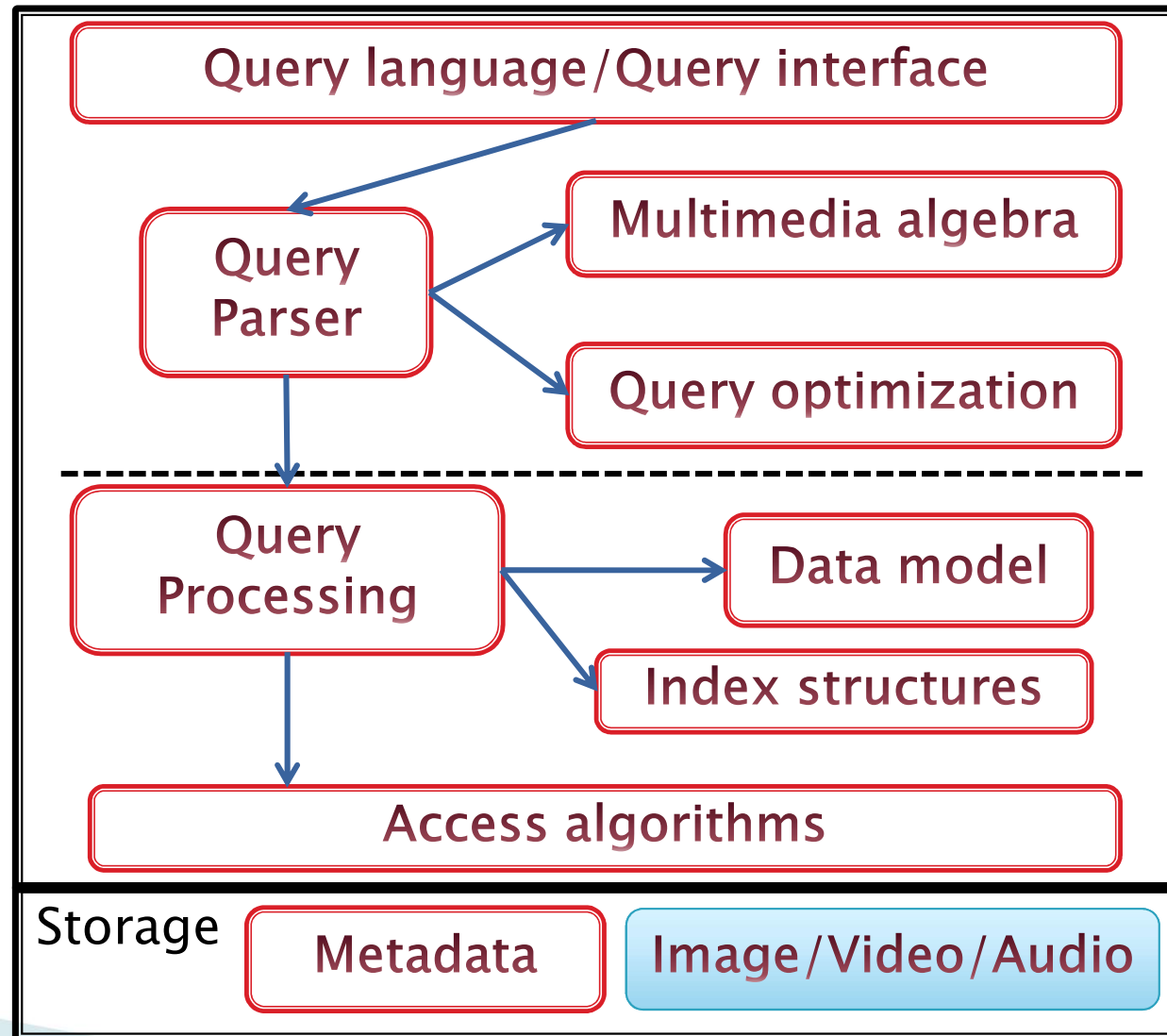
- 1 Basics
- 2 Color models

Client

Multimedia-Query

Multimedia-Data (no streaming)

Descriptive information



Multimedia Database system

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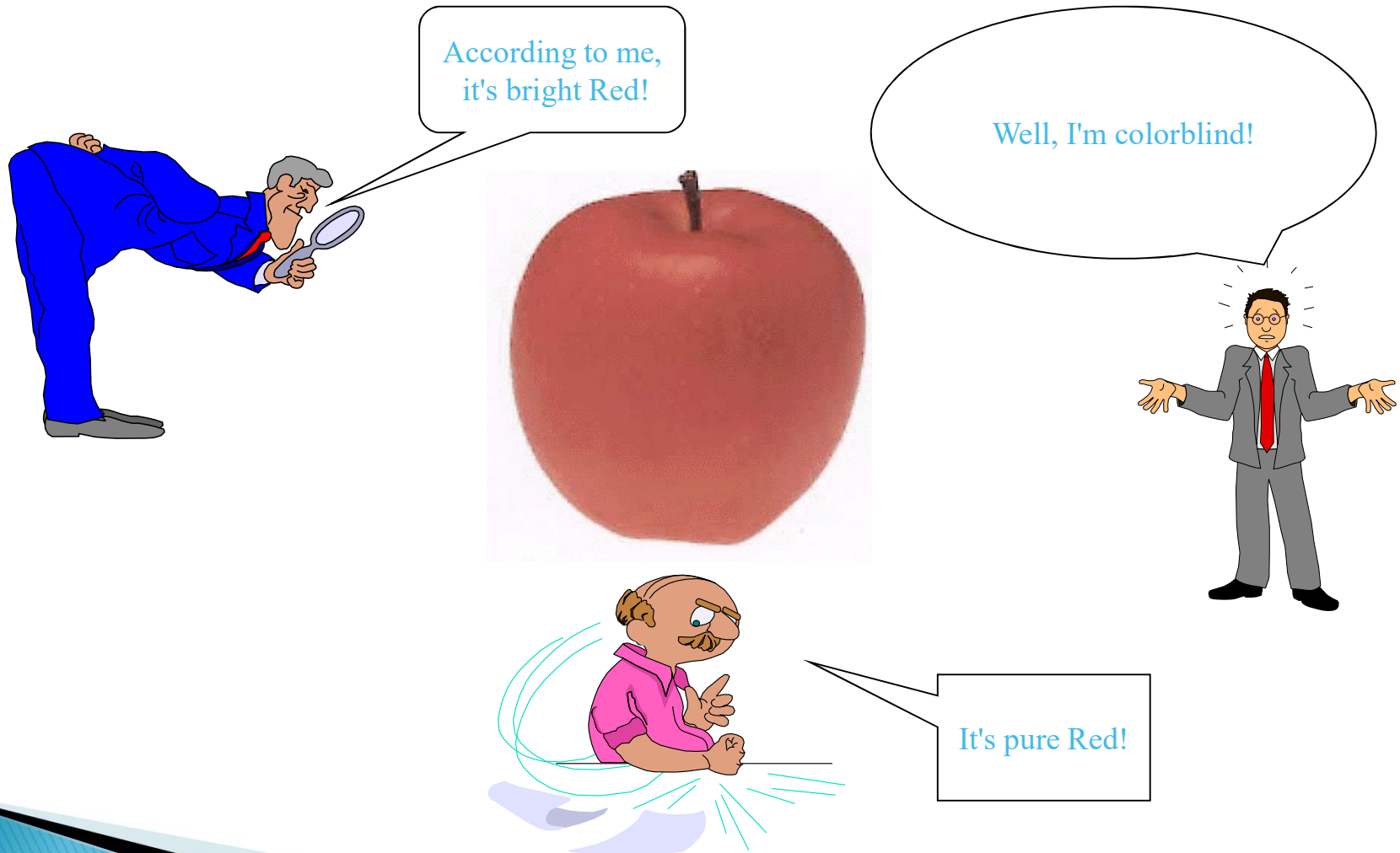
Color spaces

- 1 Color Perception
- 2 Color models



Color Perception

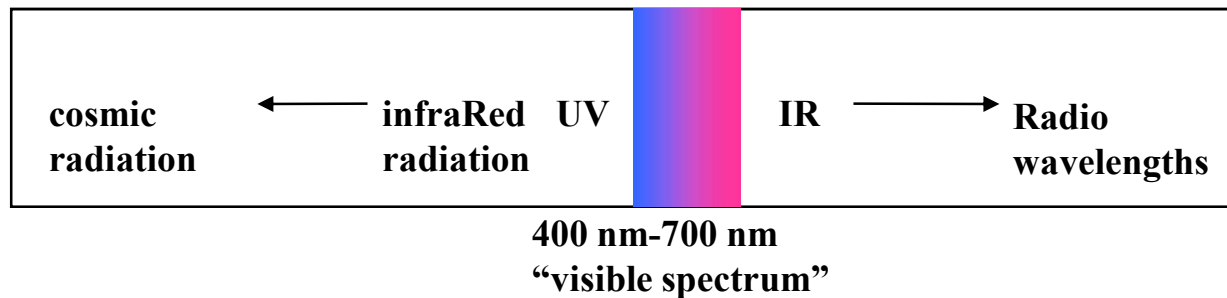
Colors: Subjective perception



Color Perception

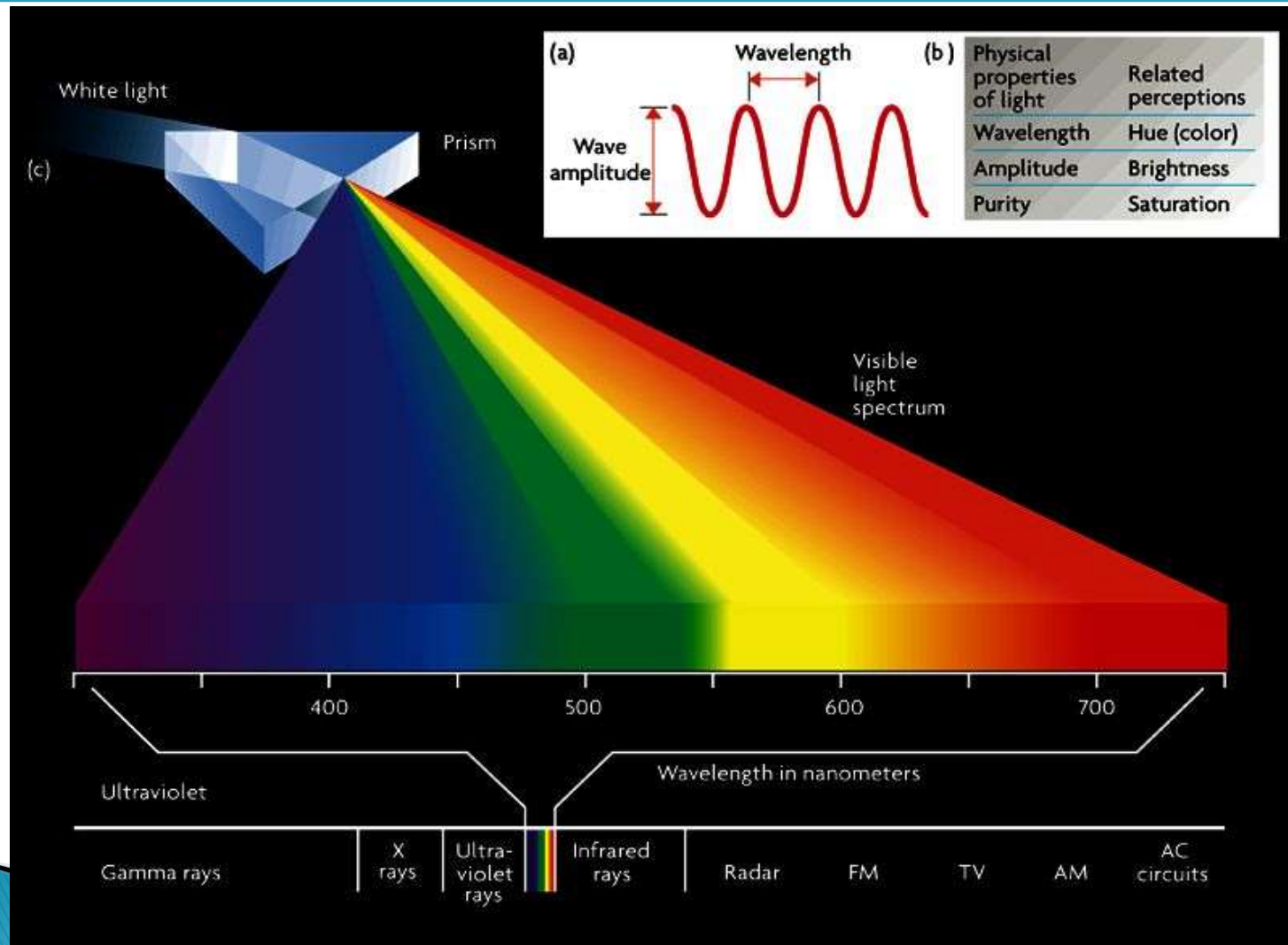
Visible light is a specific part of a large electro-magnetic spectrum

- ▶ Light is a **kind of** electromagnetic radiation
- ▶ Visible light lays between 400nm and 700nm



- Direction of light propagation
 - Straight-lined propagation
 - Scattered (ex. light bulbs)
 - Directional (ex. laser)
- Frequency
 - Color
 - Visible spectrum: 385 to 789 THz (Wavelength: 380 to 750 nm)
- Amplitude
 - Intensity of the wave (brightness)

Light is Color



Color Perception

Anatomy of the eye: 3 types of cone cells

▶ S-cones

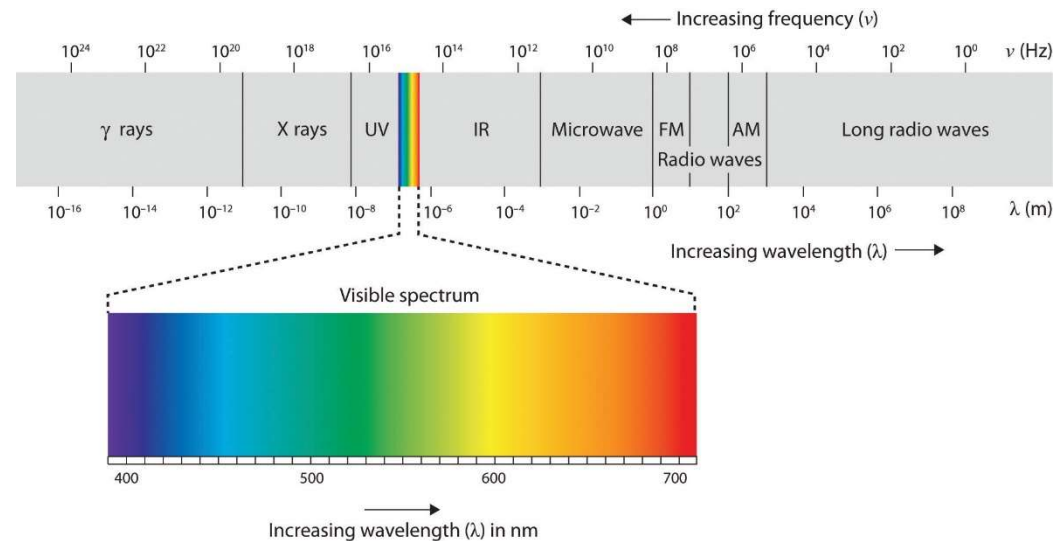
- – S: short wave length, ca. 430 nm
- – cover the blue-domain

▶ M-cones

- – M: middle wave length, ca. 530 nm
- – cover the green-domain
- – genetically closely related to L-cones
- – usual cause of color brightness

▶ L-cones

- – L: long wave length, ca. 560 nm
- – cover the red-domain



Color Perception

Luminance and Chrominance

- ▶ The eye is very sensitive to **luminance**
- ▶ 120 Million Rods and 5 Million Cones per Eye

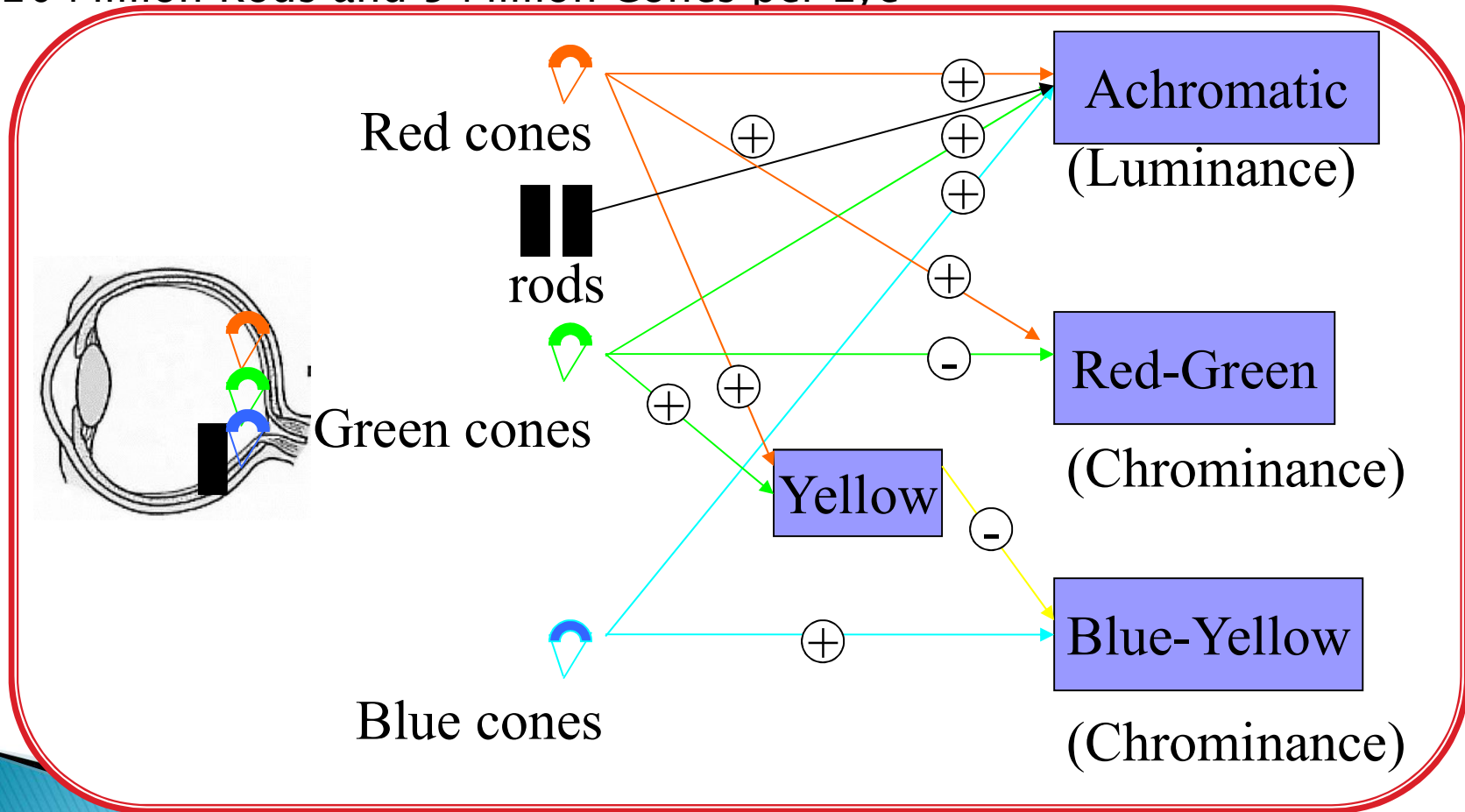


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Why color models?

- ▶ The number of spectral colors / mixes of primary colors is theoretically **infinite**
- ▶ Color models are used to capture **the widest possible part** of perceivable colors
 - Goal: **Describe reality in a simple yet relatively exact way**
- ▶ Main categories of models
 - Additive color systems
 - Subtractive color systems



Additive color

- ▶ Primary colors

- Red
- Green
- Blue
- Cyan
- Magenta
- Yellow

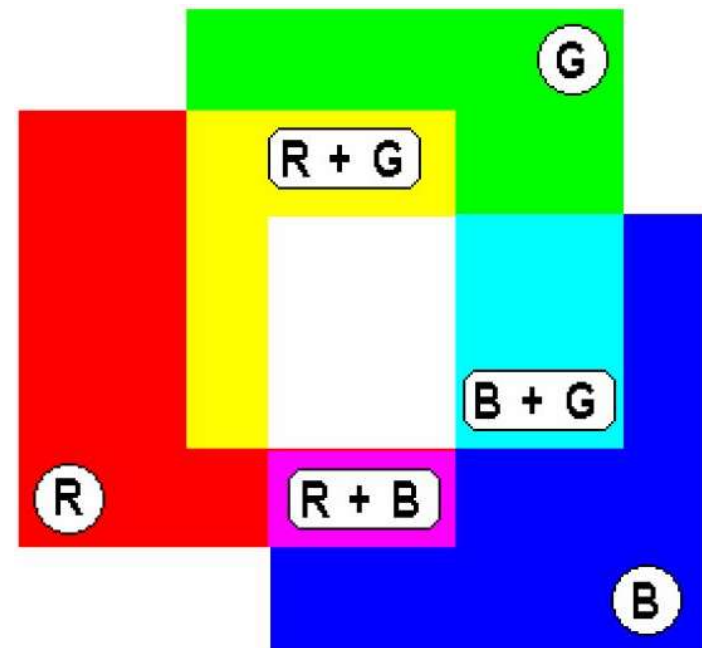


- ▶ Complementary colors

- Difference between color and white

- ▶ Color mixes

- Red + Green = Yellow
- Red + Blue = Magenta
- Blue + Green = Cyan



Subtractive color

- ▶ Primary colors

- Cyan Magenta Yellow



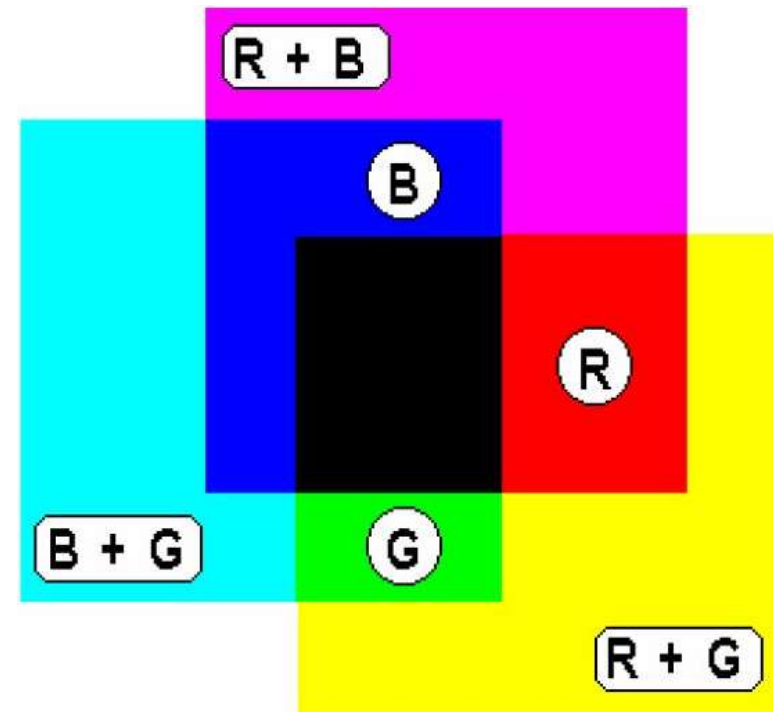
- Red Green Blue

Complementary colors

- ▶ Absorption of the complementary colour

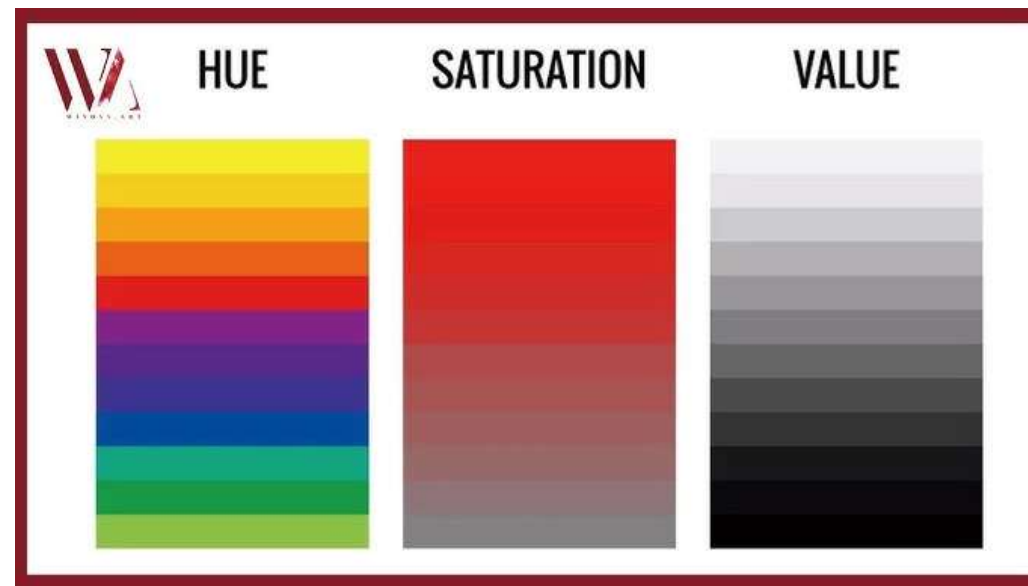
- ▶ Color mixes

- Cyan + Magenta = Blue
- Cyan + Yellow = Green
- Yellow + Magenta = Red



Hue, saturation, and luminance

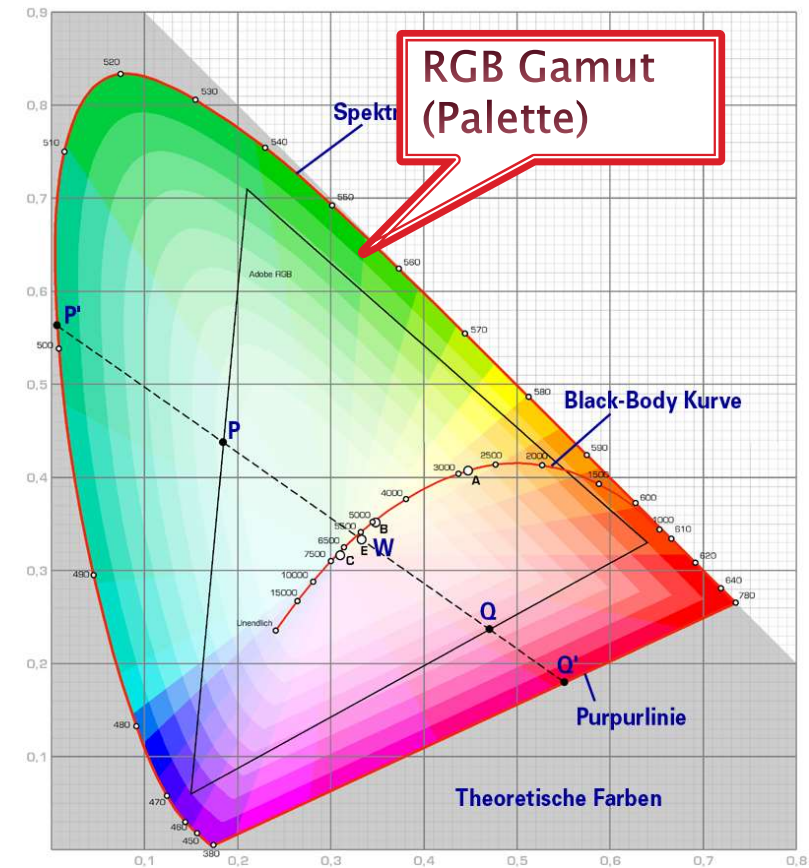
- ▶ Hue, saturation, and luminance are three key color attributes that describe how a color is perceived. Hue is the color itself, like red, blue, or green. Saturation determines how intense or pure the color is, with higher saturation being more vibrant and closer to a pure color. Luminance, also known as brightness or lightness, or value describes how dark or light a color appears, with higher luminance being brighter



Color models

CIE-Normalized color table

- ▶ Set of theoretically possible colors
- ▶ x- and y-share (from the normalized theor. primary colors X, Y and Z – computed before)
- ▶ Main condition: $x + y + z = 1$
 - then: $z = 1 - x - y$
- ▶ Reference point: **White point W**
 - Depending on illumination conditions, it can practically be located anywhere inside the horseshoe
 - **Not all colors are actually representable**
 - Here **Adobe-RGB** in the black triangle



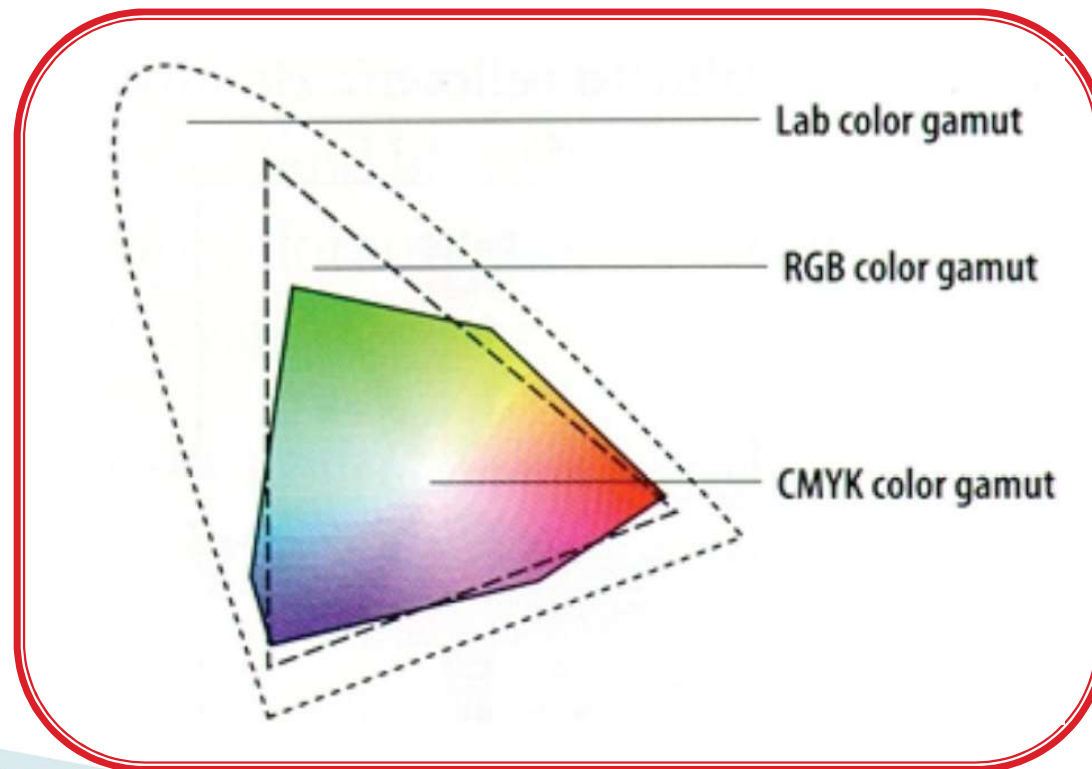
Quelle: <http://de.wikipedia.org/wiki/Bild:CIE-Normfarbtafel.png>

Diese Datei wurde unter der GNU-Lizenz für freie Dokumentation veröffentlicht.

Color models

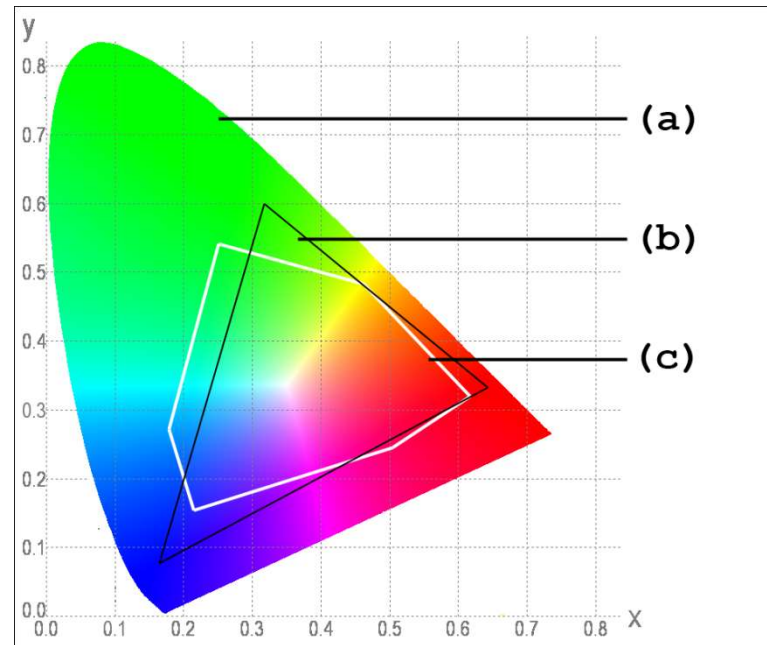
Gamut

- ▶ Domain of **colors that can be accurately reproduced by a device** (body of colors in a color space that can be produced through color mixing)



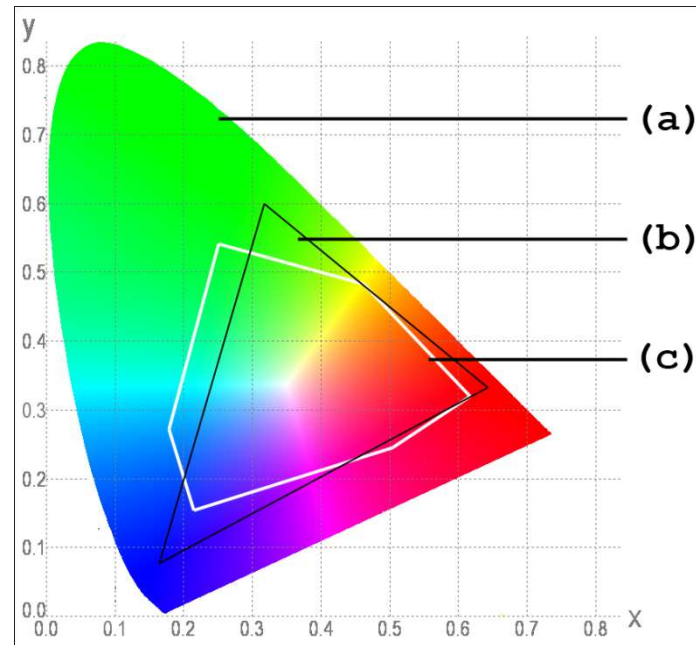
Color Gamuts

- a) Colors that human can see
- b) RGB color gamut of typical CRT monitors
- c) CMYK color gamut of typical inkjet printers



Color Gamuts

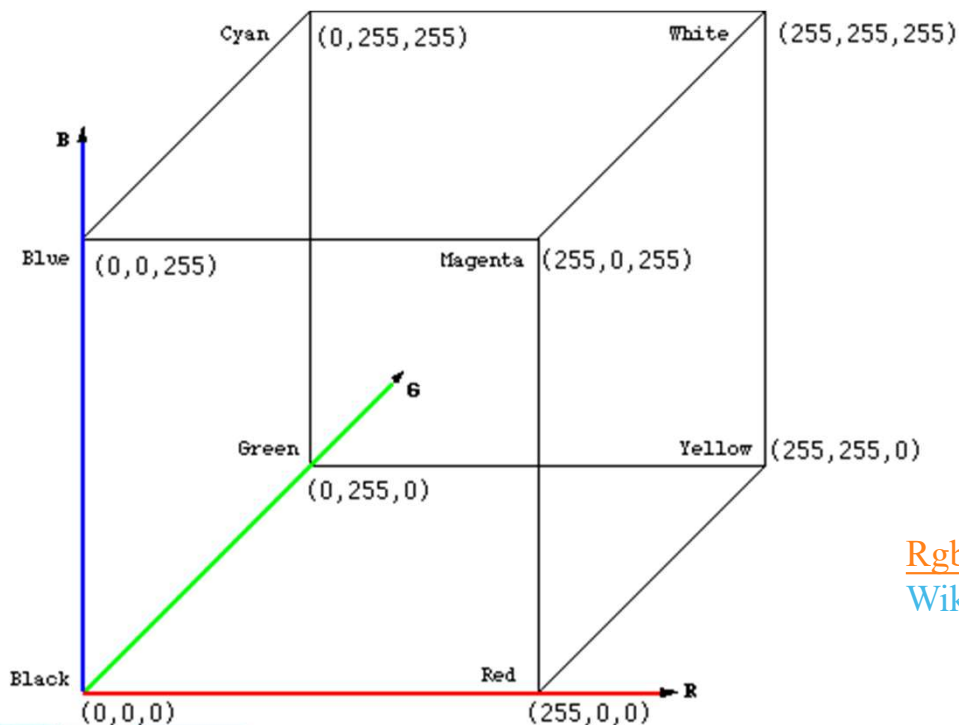
- Monitors and inkjet printers cannot reproduce all the colors that human can see
- Some of the colors that monitors can reproduce cannot be reproduced by inkjet printers. Most of these colors lie at the corners of the color gamut of the monitor, which means these are highly saturated colors.



Color models

RGB color cube

- ▶ Additive color model
- ▶ The primary colors Red, Green, Blue correspond to the **axes of the coordinate system**; they stretch to a **color cube**.

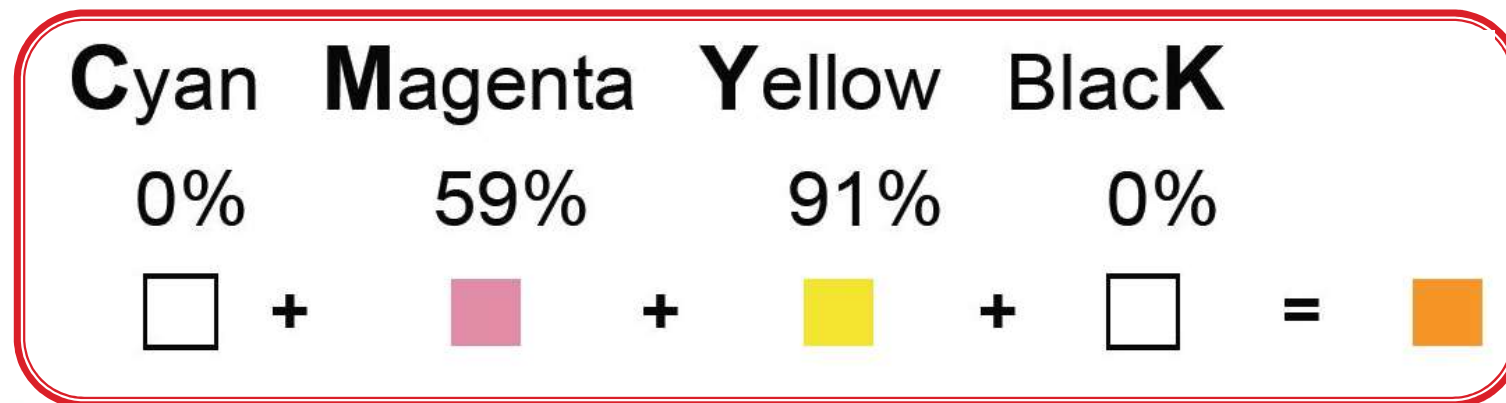


[Rgb-cube.gif](#)
[Wikimedia](#)

Color models

CMYK

- ▶ passive (subtractive) color mixing used in printing
- ▶ Four color channels: (the additional component (black) to produce dark colors)
- ▶ Technical basis of the modern four-color printing technology

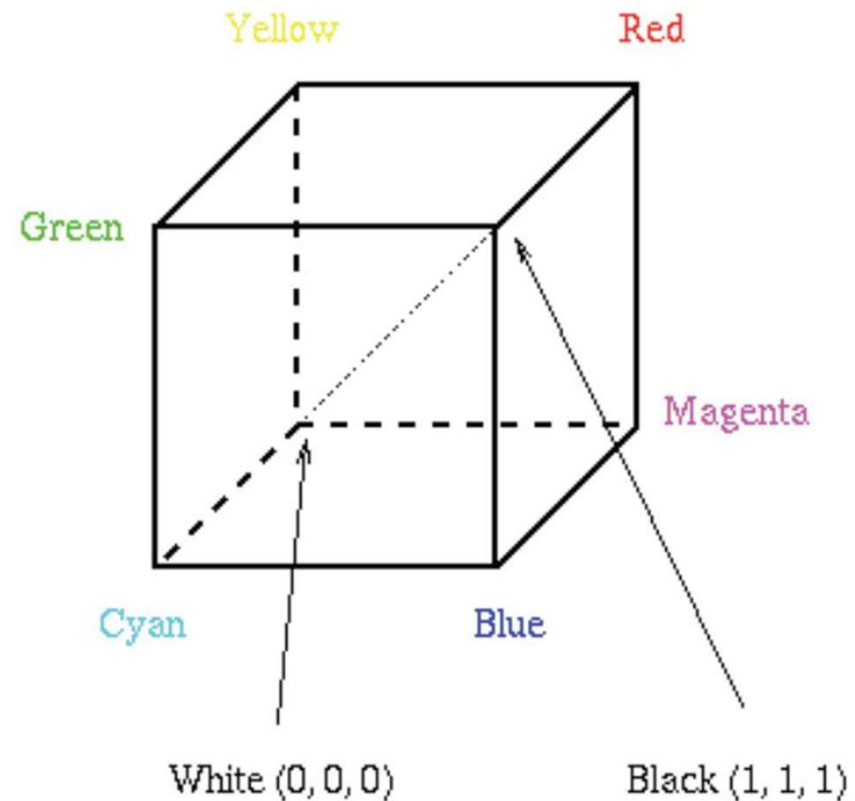


Color models

Conversion between RGB and CMY (color separation)

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

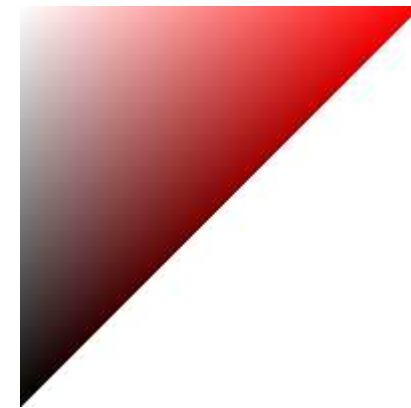
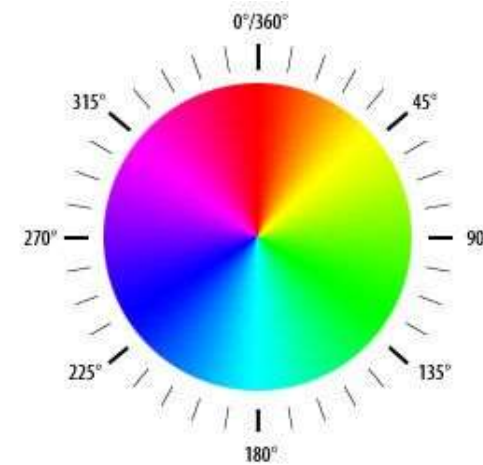
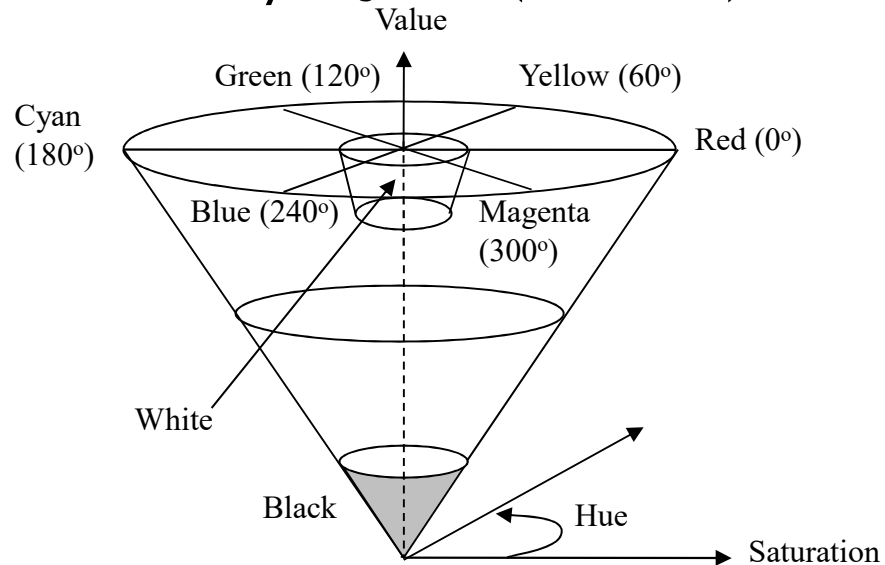
$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} C \\ M \\ Y \end{pmatrix}$$



Color models

HSB/HSV/HSI

- ▶ Hue H (Color tone) (0–360°)
- ▶ Saturation S (Color saturation) (0–100%)
- ▶ Brightness B (Helligkeit)
or Value (Darkness level)
or Intensity (brightness) (0–100%)



Color models

Converting RGB to HSV

► Conversion RGB $\rightarrow$ HSV (from Gonzalez and Woods)

1. Compute White – ratio $W = \min(R, G, B)$
2. $(R', G', B') := (R - W, G - W, B - W)$ has only 2 colorvalues $< > 0$
3. Let $B' = 0$: Hue – the value lies between R and G :

$$H = G' * \frac{120}{R' + G'}$$

Let $R' = 0$: Hue – the value lies between G and B :

$$H = B' * \frac{120}{G' + B'} + 120$$

Let $G' = 0$: Hue – the value lies between B and R :

$$H = R' * 120 / (R' + B') + 240$$

1. $S = (\max(R, G, B) - W) / \max(R, G, B)$
2. $V = \max(R, G, B)$.

Color models

HMMD-color space

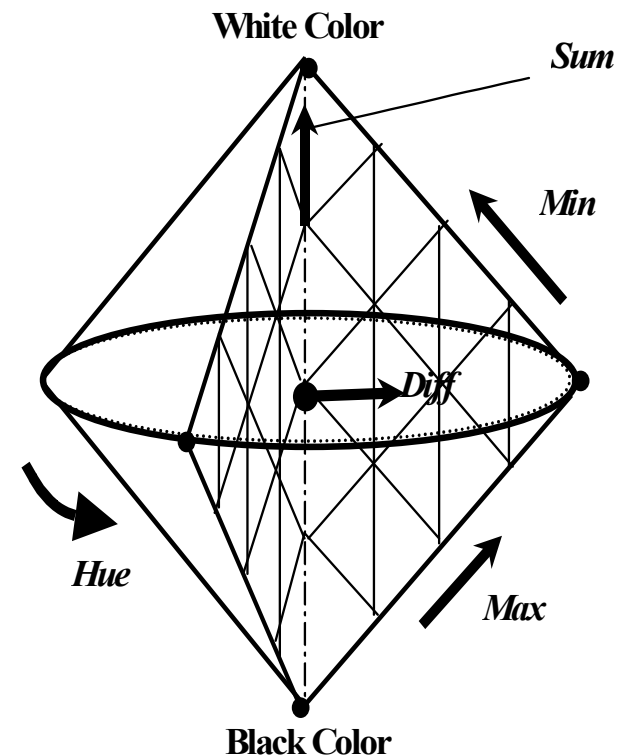
- ▶ **HMMD** (Hue–Max–Min–Diff) color space is closer to a perceptually uniform color space
 - $\text{Min} = \min(R, G, B)$; $\text{Max} = \max(R, G, B)$;
 - $\text{Diff} = \text{Max} - \text{Min}$;
 $\text{Sum} = (\text{Max} + \text{Min})/2$; Hue as in HSV

Max: gives how much "black" the color has, evaluates its darkness.

Min: gives how much "white" the color has, evaluates its pallor.

Diff: gives how much "gray" the color has and its degree of proximity to a pure color, evaluates its chrominance.

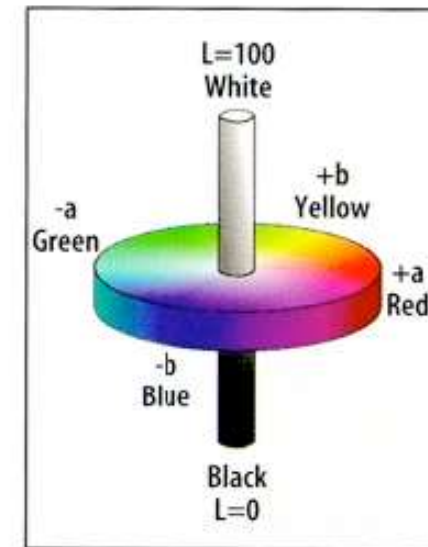
Sum: represents the brightness of the color.



Color spaces

CIE-Lab-System

- ▶ **3-dimensional color space** with the following axes:
 - L^* = Brightness (0 = black, 100 = Weiß)
 - a^* = Red – Green (-128 = Green, +127 = Red)
 - b^* = Yellow – Blue (-128 = Blue, +127 = Yellow)
- ▶ **Enhancement** of the CIE color model
- ▶ Oriented towards the **physiological properties** of human (color) perception rather than physically measurable indicators (colorimetry)



Lab model

Color models

CIE–Lab–System

- ▶ Standardized, equidistant, device independent, based on human perception
- ▶ Contains **all possible device-dependent** color spectra
- ▶ Enables **lossless conversion** from color information from a system to another



Color models

YUV/YCbCr-color space

- ▶ Y is the **Luminance** and U and V are the **Chrominance**-values of the color space
 - U (Cb): color difference Luminance – Blue, V (Cr): Color difference Luminance – Red
- ▶ **Separates** intensity and color information
- ▶ A monochrome color representation only needs the Y-value
- ▶ Model that is very close to the human perception
- ▶ Luminance can be computed from Red, Green, Blue based on physiology: $Y=0.59G+0.30R+0.11B$;
 $V=(R-Y)/1.14$; $U=(B-Y)/2.03$

Color models

conversion RGB $\rightarrow$ YUV

► Transformation

- $Y = 0,299 \cdot R + 0,587 \cdot G + 0,114 \cdot B$
- $U = 0,492 \cdot (B - Y)$
- $V = 0,877 \cdot (R - Y)$

► Reverse transformation

- $R = Y + 1.13983 \cdot V$
- $G = Y - 0.39466 \cdot U - 0.5806 \cdot V$
- $B = Y + 2.03211 \cdot U$

The end

